

Register Number

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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Computer Science and Engineering

Continuous Assessment Test – I

Regulations – R2021

Degree & Branch	B.E. & CSE				Semester	VI
Subject Code & Name	UCS2603 THEORY OF COMPUTATION					
Academic Year	2023-2024 EVEN	Batch	2021-2025	Date	15-03-2024	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Construct mathematical proofs related to computation and finite automata for a given language (K3)
CO2:	Identify relationship between finite automata and regular expressions (K3)
CO3:	Distinguish different types of grammars, analyse relationship between language and context free grammar and construct PDA for CFG (K4)
CO4:	Construct Turing machine for a given language (K3)
CO5:	Explain the decidability or undecidability of various problems and make use of it (K3)

Part – A ($4 \times 2 = 8$ Marks)

		KL	CO	PI
1.	Show a DFA which accepts all strings without two consecutive 1's over $\Sigma = \{0,1\}$.	K2	CO1	1.4.1
2.	Show the regular expression for the following: Strings of a's and b's with no consecutive a's.	K2	CO2	2.4.1
3.	What is the type of grammar given by the following set of productions? $A \rightarrow aA \mid bB$ $B \rightarrow bA \mid aB \mid \epsilon$	K1	CO3	2.1.3
4.	Find the language generated by $S \rightarrow 0S1 \mid 0A \mid 0 \mid 1B \mid 1$ $A \rightarrow 0$ $B \rightarrow 1$	K1	CO3	2.1.3

Part – B ($3 \times 6 = 18$ Marks)

		KL	CO	PI
5.	If L is accepted by an NFA then show that L is also accepted by a DFA.	K2	CO1	2.1.3
6.	Construct the regular expression whose language consists of strings with alternate 0's and 1's over an alphabet $\Sigma = \{0,1\}$.	K2	CO2	1.4.1 2.4.1
7.	State the pumping lemma for regular languages and check whether a language $L = \{ww^r \mid w \in (a,b)^*\}$ is regular or not.	K2	CO2	1.4.1 2.4.1

Part - C ($2 \times 12 = 24$ Marks)

Part - C (2 × 12 = 24 Marks)		KL	CO	PI															
8.	a. Construct a DFA for the following language L over the alphabet $\Sigma = \{0, 1\}$ (8 Marks) L = {w w when interpreted as a binary number is divisible by 3} b. Check the string validity for the number 15 using the constructed DFA. Hint: Convert 15 to required format of the model (4 Marks)	K3	CO1	1.4.1 2.1.3 2.4.1 13.3.1 13.4.1															
(Or)																			
9.	a. Construct a DFA equivalent to the NFA. $M = (\{p,q,r,s\}, \{0,1\}, \delta, p, \{q,s\})$ Where δ is defined in the following transition table. <table border="1"> <thead> <tr> <th>δ</th> <th>0</th> <th>1</th> </tr> </thead> <tbody> <tr> <td>p</td> <td>{q,s}</td> <td>{q}</td> </tr> <tr> <td>q</td> <td>{r}</td> <td>{q,r}</td> </tr> <tr> <td>r</td> <td>{s}</td> <td>{p}</td> </tr> <tr> <td>s</td> <td>-</td> <td>{p}</td> </tr> </tbody> </table>	δ	0	1	p	{q,s}	{q}	q	{r}	{q,r}	r	{s}	{p}	s	-	{p}	K3	CO1	1.4.1 2.1.3 2.4.1 13.3.1 13.4.1
δ	0	1																	
p	{q,s}	{q}																	
q	{r}	{q,r}																	
r	{s}	{p}																	
s	-	{p}																	
10.	Construct a minimized DFA for the RE $(0^*+1^*)(0+1)^*$.	K3	CO2	1.4.1 2.1.3 2.4.1 13.3.1 13.4.1															
(Or)																			
11.	Make use of Arden's Lemma to find the RE for the given DFA.	K3	CO2	1.4.1 2.1.3 2.4.1 13.3.1 13.4.1															